

CLOUD-INTEGRATED IOT AND ML-BASED WASTE MANAGEMENT SYSTEM FOR RAILWAY STATIONS

25-26J-191

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology Specialized in Information
Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

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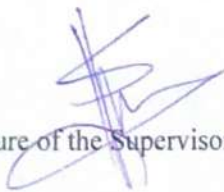
August 2025

DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.


Signature of the Supervisor:

24/8/25
Date:

Abstract

Handling waste effectively in bustling public areas is a persistent issue, especially in settings like railway stations where waste patterns are irregular and unpredictable. The conventional waste management practices are highly labor-intensive and have led to premature irregular cleaning schedules, lag in waste removal, and hence higher operational costs. To overcome these limitations, this research seeks to create a standalone garbage detection and collection system that can operate without human intervention but delivers accuracy and efficiency. The system being developed uses a combination of ultrasonic sensors, a camera, a classification model using machine learning, and a robotic arm on a mobile robot platform. The process begins with ultrasonic sensors on or near the ground to detect if there is a potential waste object within the robot's collection range. The camera then photographs the detected item and feeds it into a machine learning model specially designed to identify if the object is waste or not. This classification step ensures that the system activates the robotic arm only when an actual waste item is confirmed, preventing unnecessary energy consumption and mechanical wear. If it identifies the object as trash, the robotic arm gently lifts the item and places it inside the robot's onboard trash bin, preparing the system for the next detection and pickup cycles. With real-time object detection, AI-based object classification, and mechanical automation, this system offers an optimal, selective, and scalable waste management solution for public spaces. The innovation is combining different sensing and decision-making techniques powered by artificial intelligence to create a robust, low-maintenance, and adaptive waste collecting system that minimizes human intervention to a considerable extent while providing continuous cleanliness for railway stations.

Keywords: Autonomous waste collection, Garbage detection, Machine learning, Object classification, Robotic arm

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LIST OF ABBREVIATIONS

Abbreviation	Description
IoT	Internet of Things
AI	Artificial Intelligence
SLAM	Simultaneous Localization and Mapping
IR Sensor	Infrared Sensor
GPS	Global Positioning System
IMU	Inertial Measurement Unit
QA	Quality Assurance
IDE	Integrated Development Environment
MS Planner	Microsoft Planner
SIFT	Scale-Invariant Feature Transform
HOG	Histogram of Oriented Gradients
SVM	Support Vector Machine
CNN	Convolutional Neural Network
R-CNN	Region-based Convolutional Neural Network
YOLO	You Only Look Once
ML	Machine Learning

1. INTRODUCTION

1.1 Background & Literature Review

Proper management of garbage in public places such as streets, transport stations, and markets has been recognized as a serious urban issue for decades. Conventional approaches rely on stationary bins and hand-sweeping, which rarely address the dynamic and unpredictable nature of waste disposal in such environments. Litter lying en route and over-flow bins not only decrease the cleanliness of public spaces but also pose health hazards and increase passenger dissatisfaction. These problems have fueled interest in autonomous robotic platforms that are able to sense, collect, and dispose of waste without need for ongoing human intervention [1], [2], [9], [10]. By integrating sensing technologies, computer vision, and robotic manipulation, these systems are attempting to bypass the inefficiencies of human cleaning and offer scalable and dynamic solutions to dynamic conditions like that of railway stations.

Real-time identification and classification of trash objects are a primary capability of these robots. Computer vision supported by deep learning has emerged as the foundation of modern waste detection systems. The initial object detection techniques relied upon the application of handcrafted features such as SIFT and HOG coupled with classifiers like SVMs. Deep convolutional neural networks (CNNs) and top-of-the-line detectors like YOLO (You Only Look Once) [4] and Faster R-CNN [3] dominate the scene now due to their relative speed-accuracy tradeoff. YOLOv3 [1], YOLOv5 [4], and YOLOv7-based systems [5] have been found to be highly accurate in detecting non-regular-shaped objects such as cans, bottles, and wrappers even in messy and complex environments. Domain-specific waste detection models also enhance their robustness by using domain-specific datasets as compared to general-purpose image datasets, a huge leap for real-world waste collection processes [7], [8].

In addition to computer vision, proximity sensing technologies provide key help in distinguishing between waste items and other environment. Ultrasonic sensors are particularly beneficial in differentiating between small floor-level litter and tall objects such as walls, benches, or human legs. In the case of trash collection using a robot,

ultrasonic sensors at varying heights can trigger the vision system to operate only when potential trash is detected, with the aim of reducing computation overhead [2], [6]. Hybrid sensing solutions grounded in deep learning combined with ultrasonic or infrared sensors [7] have been shown to improve system robustness under conditions with changing illumination or occlusion, for example, railway stations.

Detection alone is not sufficient; robots must also collect trash effectively. Robotic arms and grippers have been thoroughly explored for autonomous trash collection. Low-degree-of-freedom manipulators are normally preferred on account of simplicity of use, low cost, and suitability for picking lightweight trash fragments [3]. Prototypes of robot arms integrated with waste identification systems have already demonstrated practicability for grasping objects and putting them inside onboard containers [1], [3]. However, difficulty arises with manipulations of irregular shapes, slippery surfaces, and intertwined object orientations that are apt to cause grasping failure. Studies on adaptive grasping methodologies, like feedback-based robotic manipulation, report better performance in waste collection [3], [7].

Recent studies also emphasized the necessity to integrate detection, classification, and collection into faultless autonomous systems. Robots like those in [1], [3], and [7] couple CNN-based trash detection with robotic arms to capture and deposit trash in real-time. These systems are typically restricted in scope, however, and are meant for specific narrow types of trash or small spaces. Advanced work has attempted to enhance waste classification granularity by distinguishing between types such as plastics, paper, glass, and polythene to correspond with real recycling processes [5], [8]. YOLOv7-based [5] classification has provided promising outcomes in improving detectability for non-recyclable and recyclable classes in real-world scenarios.

Performance also relies greatly on the working environment. Train stations, unlike indoor controlled environments, present specific challenges of high pedestrian crowds, frequent occlusion of trash objects, and varying illumination. The models such as YOLOv5 [4] and YOLOv7 [5], which are lightweight deep learning models, are proposed to operate on embedded hardware to undertake real-time inference in such resource-constrained environments. Such models provide low-latency detection with

the ability to maintain efficiency, a requirement for robots to operate independently on busy train stations.

Despite these advances, there are still limitations. The majority of cutting-edge systems still rely on diminished assumptions such as unobstructed views of garbage or homogeneous lighting conditions. Trash detection in crowded, outdoor, or partially structured environments such as train stations remains an open research problem. Moreover, despite advances in robotic grippers, flexibility in handling non-standard or occluded shapes of garbage is still restricted. Cloud integration, multi-robot coordination, and IoT-monitored surveillance key features of modern smart systems are often lacking from current prototypes [2], [6].

Generally, reviewed literature shows significant developments in autonomous collection and detection of waste by the use of deep learning, sensor fusion, and robots for manipulation. Robotic arms are aided to facilitate physical picking, although there still remains challenge with irregular shapes and clutter. CNN object detectors such as YOLOv3, YOLOv5, and YOLOv7 allow accurate identification, while ultrasonic sensors allow distinction between solid objects in real-world scenarios. Current prototypes demonstrate feasibility but lack waste category scope, versatility, and IoT integration. Closing these gaps with enhanced datasets, adaptive grabbing tactics, and lightweight embedded deep learning models can provide the pathway to deployable autonomous garbage-gathering robots in railway terminals and other heavy-traffic zones.

1.2 Research Gap

The robotization of waste detection and collection has been explored extensively in service robotics and smart city applications. Existing approaches commonly employ stationary smart bins or semi-autonomous robots that rely on basic sensing mechanisms such as ultrasonic distance measurement and rule-based detection [2], [6]. While these systems demonstrate feasibility, they often depend on overly simplistic strategies, including threshold-based waste recognition or broad classifications that divide garbage into only recyclable and non-recyclable categories [5], [8]. Such limited frameworks fail to reflect the complexities of real-world environments, particularly in

crowded railway stations where waste objects vary drastically in shape, size, and material, and must be detected under variable lighting and highly dynamic conditions.

Recent computer vision research highlights the capability of deep learning models such as YOLOv3, YOLOv5, and YOLOv7 in waste detection and classification tasks [1], [4], [5]. These models demonstrate high detection accuracy but are often trained on generic or laboratory datasets, leading to reduced performance when transferred to real-world railway settings where contextualized training is essential. Similarly, robotic manipulators have been extensively investigated in industrial domains [3], but their optimization for deformable, irregular, and lightweight waste objects remains underdeveloped. The integration of perception and actuation where a robot not only detects waste but also successfully grasps and deposits it into an onboard container is still insufficiently addressed in the literature [1], [3].

Another major gap lies in the absence of end-to-end frameworks that unify real-time sensing, deep learning-based classification, and robotic manipulation into a single autonomous pipeline. Studies on smart garbage robots such as those utilizing ROS-based systems [2] and computer vision-guided detection [7] provide partial autonomy but often rely on simplified environments, pre-defined detection rules, or limited waste categories. More advanced models, such as improved YOLOv7 architectures [5], indicate the potential for fine-grained garbage classification but remain disconnected from robotic collection in dynamic contexts. While several prototypes, including YOLO-powered garbage detection [1], [4], [7], have achieved significant progress, they do not yet extend to real-time synchronization between detection and robotic arm response in railway environments, where continuous occlusion and passenger density introduce additional complexity.

Therefore, the research gap is centered on the lack of comprehensive, context-aware, end-to-end waste collection systems capable of performing reliable detection, classification, and manipulation in railway stations. This underscores the need for lightweight yet accurate deep learning models integrated with adaptive robotic manipulators to ensure efficient and autonomous garbage management in such semi-structured, high-traffic environments.

Table 1.1 Comparison of previous researches

Existing System	Approach Used	Strengths	Gaps Identified
Garbage Collecting Robot Using YOLOv3 [1]	YOLOv3 deep learning model for real-time garbage detection and robotic collection	Achieves real-time detection of waste items with higher accuracy compared to traditional methods	Limited dataset training; performance decreases in cluttered and dynamic public environments such as railway stations
ROS in Smart Garbage Bin Robot [2]	Robot Operating System (ROS) integrated with smart bin and IoT connectivity	Provides modular design, efficient ROS-based integration, and IoT monitoring	Limited to stationary smart bins; does not address mobile, real-time collection in crowded spaces
Intelligent Robot for Garbage Collection and Compression [3]	Autonomous garbage collection with robotic arm and onboard compression mechanism	Introduces waste volume reduction, integrating detection with robotic manipulation	Focused on compression efficiency; lacks optimized detection for deformable and irregular waste objects
Using YOLOv5 for Garbage Classification [4]	YOLOv5-based waste classification with improved accuracy	Achieves better classification speed and precision compared to YOLOv3	Tested mostly in controlled datasets; not validated in real-world railway-like environments
Garbage Classification using YOLOv7 [5]	Improved YOLOv7 for garbage detection and fine-grained classification	Provides more accurate classification into multiple categories	Not coupled with robotic manipulation; does not address full detection-to-collection pipeline

1.3 Research Problem

How can autonomous mobile robots be designed to effectively detect, recognize, and collect different waste objects in complex, cluttered environments such as railway stations by combining vision models, sensor fusion, and robotic arm control, and what are the implications of deploying such systems on large-scale public trash management?

One of the biggest persisting problems in modern waste management is the detection and collection of garbage in active public areas such as train stations. Unlike laboratory settings with controlled environments, train platforms present random conditions: variable lighting, high foot traffic, diverse forms of trash, and constantly changing object locations. Traditional cleaning systems rely on human effort, which is slow, labor-intensive, and often inconsistent in maintaining sanitation levels. While there have been some robotics solutions suggested, they are either limited to simple obstacle detection or static garbage collection, and few have been able to integrate intelligent waste identification with robotic autonomous collection.

The problem is to enable a mobile robot to not just find trash objects accurately but also to separate them from background items and undesired material. Vision models and cameras provide one level of detection, although these are prone to environmental variations such as lighting, shadows, and occlusions. Similarly, ultrasonic sensors can provide the presence of an object but not if the object is waste or not. Coupling the detection system with a robotic arm for physical collection is another complexity since wastes differ in size, shape, and position, which requires adaptive manipulation techniques.

2. OBJECTIVE

2.1 Main Objectives

The main objective is to develop a cloud-integrated, IoT- and machine-learning-based autonomous waste management system that can detect, collect, classify, and intelligently unload garbage in railway stations, while ensuring real-time monitoring of disposal bins and timely notifications for efficient waste handling.

2.2 Specific Objectives

- Autonomous Garbage Detection and Collection
 - To design and implement a vision-based detection mechanism supported by sensors and a robotic arm for identifying and collecting garbage items in real time.
- Intelligent Navigation and Unloading Control
 - To enable the mobile robot to autonomously navigate railway station environments with obstacle and fall prevention, continuously monitor internal bin fill levels, and perform automated unloading at predefined disposal areas.
- Real-Time Crowd Detection and Waste Classification
 - To develop and train custom machine learning models for classifying collected waste into four categories as Plastic, Paper, Glass, and Polythene the disposal station, and detecting crowd density through head-count analysis using cameras to ensure safe navigation and operation.
- Automated Waste Sorting and Disposal Bin Monitoring
 - To integrate a robotic arm for placing classified waste into static bins and implement IoT-based monitoring of disposal bin fill levels, with cloud-based visualization and real-time notifications for cleaning staff.

3. METHODOLOGY

3.1 System Architecture

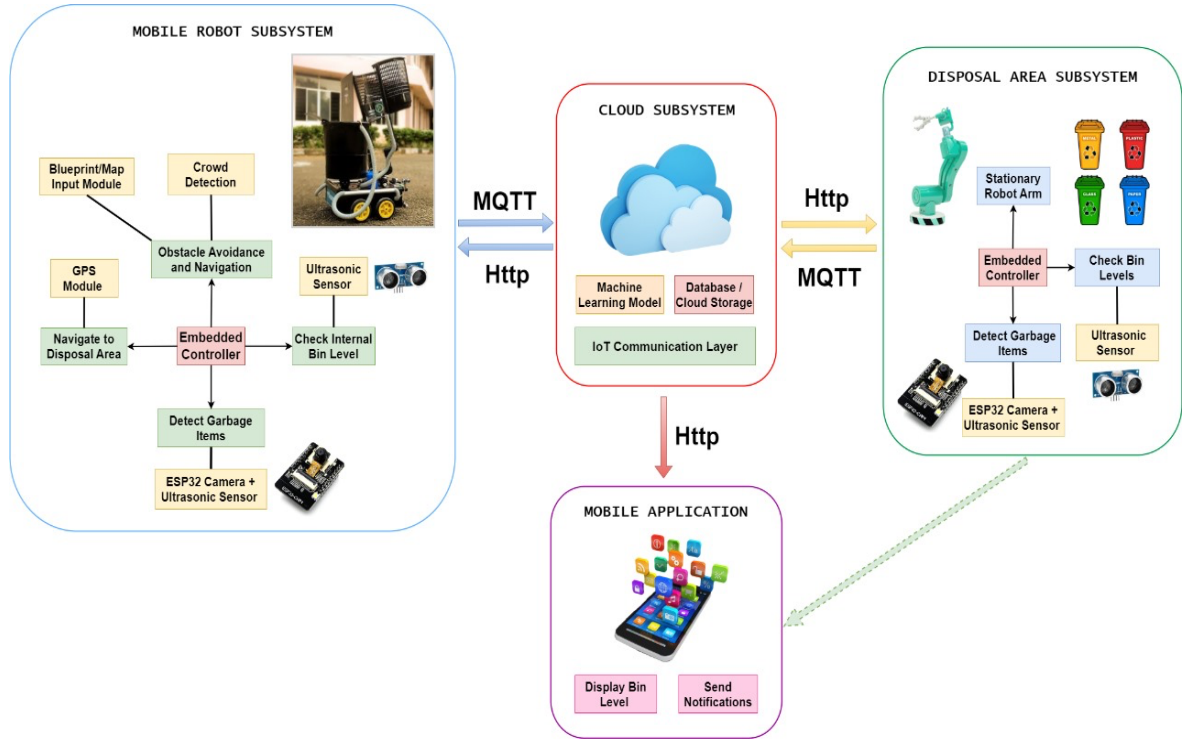


Figure 3.1: System architectural diagram for the overall system

Figure 3.1 presents the overall system architecture, illustrating how the four subsystems mobile robot, cloud, disposal area, and mobile application are interconnected to achieve seamless autonomous waste detection, collection, classification, and disposal in railway stations.

The system has been conceptualized as an integrated waste management system which utilizes robotics, IoT, and machine learning to make the railway stations cleaner. It has been segmented into four standalone subsystems that work in flawless synergy to create a fully autonomous, efficient, and reliable system for waste collection,

segregation, and disposal. Each of the subsystems offers an individual set of capabilities, but how they interoperate is what results in the achievement of the overall research objectives.

- **Mobile Robot Subsystem:** This unit offers the essence of the solution by serving as the active unit for performing the process of waste collection. It has an onboard vision system supplemented by sensors that help identify garbage objects in real-time. Upon detection of an object, the robotic arm switches on automatically to pick up the trash and deposit it into the robot's internal storage compartment. The navigation module allows the robot to move freely through station environments without facing hazards and obstacles. Safety is ensured with the use of sensors placed at strategic positions to not only navigate objects around in its path, but also to avoid falling off elevated railway platforms. In addition, the subsystem integrates blueprint-based spatial awareness to pre-identify static obstacles, safe zones, and platform edges, enhancing navigation safety. The robot is also aided by a crowd detection mechanism via cameras and a learned model to stop navigating and move towards the closest predetermined safe spot upon detection of intense crowding, resuming operation once crowds have abated. The subsystem further possesses a vigilance system that checks the fill level of its internal tank constantly to offload only as needed. After filling, the robot moves automatically to the disposal point with the help of location awareness and GPS.
- **Cloud Subsystem:** The cloud subsystem provides central intelligence and connectivity. It is the site of the machine learning algorithm that translates picture data to waste classification to deliver accuracy in the discrimination between types such as plastic, glass, paper, and polythene. Apart from classification, the cloud gives the communications infrastructure, connecting the mobile robot, disposal site, and mobile application into a unified architecture. It also addresses data storage, with the ability to conduct long-term trend analysis of waste and scalability to scale the solution to numerous railway stations. Additionally, the cloud hosts the crowd detection model for

head-count analysis, transmitting results to the robot in real time to guide safe navigation and trigger safe-zone protocols when necessary.

- **Disposal Area Subsystem:** In the disposal area subsystem, waste handling changes from collection to segregation. Once the mobile robot off-loads the internal bin onto the platform, it is passed over by a fixed robotic arm with inbuilt controllers. This arm, which is guided by machine learning classification output, sorts the collected items into dedicated bins. The disposal bins all feature ultrasonic sensors that constantly track and report fill levels. By embedding intelligence in this component of the system, the disposal is systematic and sustainable, reducing human interference and mistake in segregation of waste.
- **Mobile Application:** The mobile interface is the human-machine interface of the system. The disposal systems and the robot run independently, but the mobile app provides visibility and accountability. It displays actual fill levels for disposal bins, giving cleaning staff complete visibility of the system's status. Automatic notifications are issued when a bin is close to full capacity, prompting timely action. This ability closes the loop of autonomous and human monitoring, always remaining effective in protecting railway station sanitation.

Through the integration of the foregoing subsystems, the solution responds to several concerns simultaneously: autonomous garbage detection and pickup, free passage in crowded public areas, smart separation and sorting, and effective human-machine interaction. The combination of IoT, cloud-based intelligence, and robotics enables the railway stations to dynamically maintain cleanliness levels based on actual conditions rather than predetermined schedules. Coupling autonomy with intelligent notifications and real-time monitoring, the system offers a revolutionary approach to modern waste management of high-density public spaces.

3.2 Process Workflow

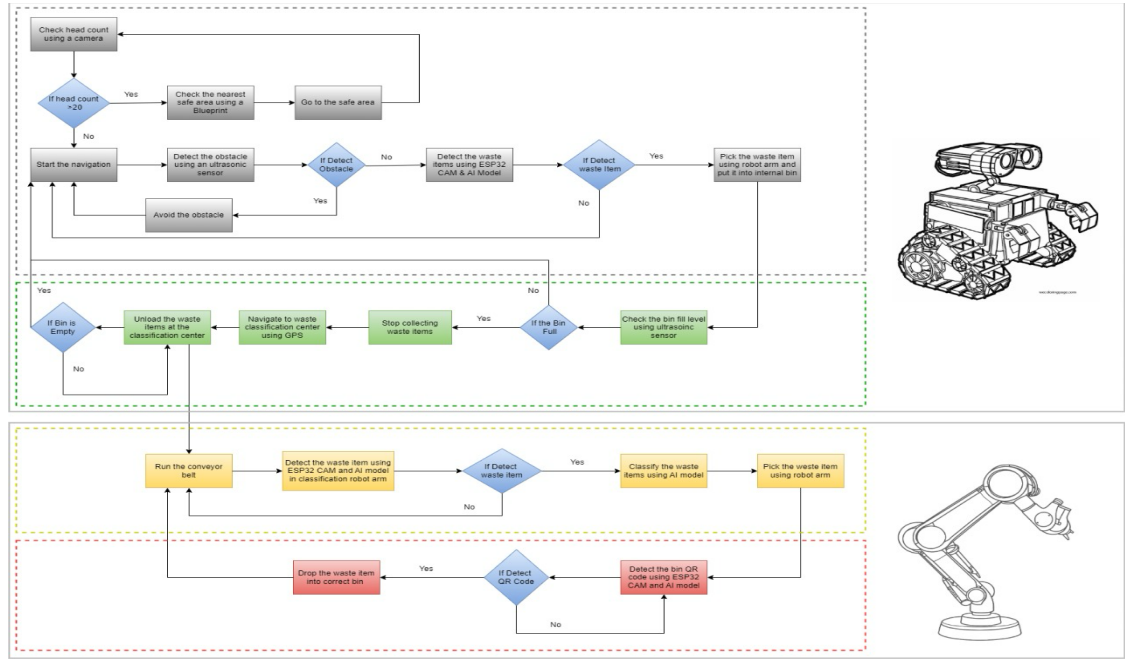


Figure 3.2: Flowchart for the overall system

The overall workflow of the proposed system has been designed to ensure that waste is identified, collected, transported, and classified autonomously and effectively with real-time monitoring and alert. The system begins with the autonomous robot actively navigating around the area to sense waste objects. According to onboard sensors and an in-built vision model, the robot separates potential garbage objects from other irrelevant environmental objects. After identifying a valid garbage object, the robotic arm is activated to grasp the object and deposit it into the robot's internal bin.

As the robot continues operation, the internal bin level is continuously monitored by ultrasonic sensing. When the bin is 80% (or a predetermined percentage) full, the system proceeds to the unloading phase. In this stage, the robot switches its task from active collection to navigation. The robot navigates to the disposal point via a combination of location awareness and obstacle avoidance so that the robot can safely operate within congested or semi-structured railway station environments. To enhance navigation, the robot also uses blueprint-based spatial awareness to pre-identify static obstacles, platform edges, and predefined safe zones. Furthermore, a crowd detection

mechanism using camera input and a trained model monitors real-time head counts. If a crowded situation is detected during operation, the robot halts its collection task and moves to the nearest predefined safe zone, resuming only when congestion levels decrease.

Once the robot reaches the disposal area, it performs the unloading process, emptying collected waste onto a dumping stage. The disposal system is also triggered where the classification model deposits every item of waste into four bins named Plastic, Paper, Glass, and Metal. Each product is released by the fixed robotic arm into its designated dump bin, which has been preset up and placed in prearranged locations for standardization.

Simultaneously, all the garbage bins are equipped with ultrasonic sensors to constantly monitor their levels. These are sent via the IoT layer to a cloud database. A mobile application makes this visual, providing cleaning staff with real-time feedback and notifications when a bin is reaching full capacity. This enables timely human intervention only when required, thereby avoiding spillage and unnecessary checking.

Figure 3.2 illustrates this end-to-end process, beginning with initial trash detection, followed by navigation and unloading, and concluding with final classification and notification. This structured workflow illustrates how multiple subsystems come together in unison to achieve the overall goal of autonomous waste management in railway stations, while ensuring crowd-aware safety, blueprint-based navigation accuracy, and efficient bin monitoring

3.3 Individual Component

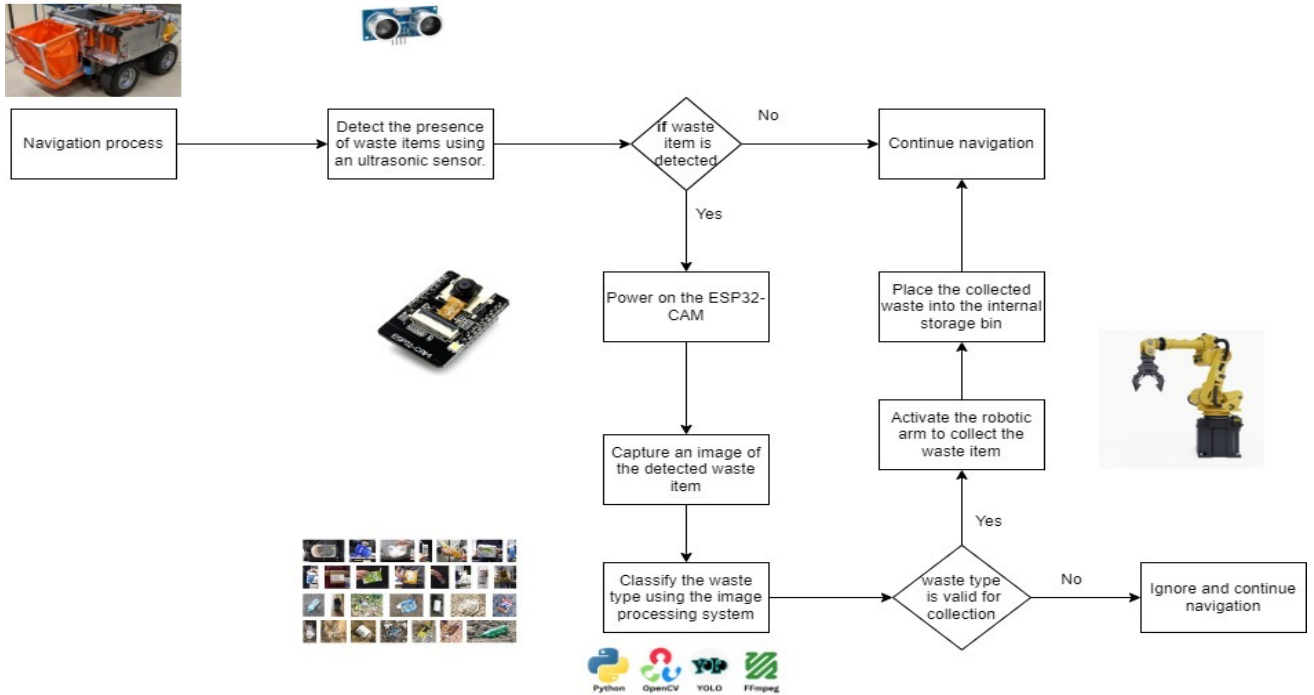


Figure 3.3: Individual component diagram

The autonomous garbage detection and collection module is the first stage of the smart waste management system that is responsible for detecting, capturing, and storing safely the garbage objects in the inner bin of the robot. The module enables the robot to directly engage with the environment through sensing and physical manipulation of waste, setting the stage for the subsequent navigation and disposal processes.

It begins by sweeping the environment to identify potential waste objects. Proximity sensors installed close to the ground help detect objects on the floor. When the system detects a trash item, it initiates a vision-based verification stage through the use of a camera that captures an image of the object. The image is processed through the use of a trained classification model to confirm whether the object detected is trash. By combining sensor-based detection with visual confirmation, the system avoids false positives and targets only the items of interest for collection.

When detected and confirmed, the robotic arm mechanism is triggered to perform the collection process. The arm is set to automatically reach the detected object, precisely grasp it, and lift it into the robot's inner bin. The collection process is performed autonomously without human intervention, enabling the robot to continue detecting and extracting trash objects from the environment.

The novelty of this unit lies in the seamless integration of autonomous sensing, visual recognition, and robotic manipulation in a railway station setting. As opposed to traditional systems that rely on manual pickup or rudimentary sweeping machines, this approach allows the robot to intelligently identify and physically collect different types of trash. This combination of detection and robot action is the foundation for a fully autonomous garbage collection cycle, reducing the necessity of human cleaners and enhancing sanitation effectiveness in high-traffic rail environments.

3.4 Project Management and Development Approach

3.4.1 Requirement Identification

The development begins with a clear understanding of how garbage is scattered across railway stations and what challenges exist in detecting and collecting it. Inputs are gathered through observation at stations, reviewing prior robotic waste collection studies, and consulting cleaning staff to understand the nature of commonly found waste. The outcome is a validated list of requirements that focus on accurate detection of garbage items and reliable robotic arm operation for collection.

3.4.2 Technology Decision

Hardware and software platforms are selected with emphasis on balancing cost, accuracy, and ease of implementation. The detection unit is designed around a combination of ultrasonic sensors for object presence and a camera module linked with a trained ML model for waste identification. Lightweight actuators are chosen for the robotic arm to ensure precise but power-efficient collection.

3.4.3 Iterative Development Cycles

The work is divided into multiple small development cycles. Each cycle delivers a part of the functionality such as initial waste sensing, image capture and classification, or

arm actuation for pickup. This phased approach ensures that each feature can be tested and refined independently before integration into the larger robot framework.

3.4.4 Testing and Feedback

After every development cycle, functions that have been implemented are validated in simulated environments replicating real-world environmental scenarios. For example, different kinds of waste are let fall in front of the robot to validate detection accuracy and ability of the arm to grasp them. Test outcomes are utilized to adjust the trained model and alter arm control logic.

3.4.5 Integration and Validation

When detection and collection modules operate independently, they are inserted into the mobile platform. Continuous testing allows smooth coordination of sensors, camera, and robot arm. During this stage, lifting capacity of the arm, up to **a maximum of 2 kg**, is also tested to ensure safe and effective grasping of regular waste items. This phase ensures that trash is detected properly and routed to onboard bin without failure.

3.4.6 Maintenance and Updates

Since the machine learning model may need updates as new datasets are collected, a plan is set for periodic retraining and re-deployment. Mechanical wear of the robotic arm is also considered, with guidelines prepared for recalibration, servicing, and verification of its lifting capacity over time.

3.4.7 Reliability Assurance

To ensure robustness, multiple tests are carried out across different lighting and cluttered environments. Accuracy of detection, consistency of the robotic arm grip, and timing of operations are carefully validated to confirm that the system achieves dependable real-time garbage collection.

3.5 Hardware, Software, and Resource Methodology

3.5.1 Hardware (Sensing and Robotic Manipulation)

The hardware element of this unit is responsible for sensing trash items, monitoring them, and physically collecting them into the onboard bin. The decision has been made

with an emphasis on cost, accuracy, and durability of utilization in real railway environments.

Sensors

- **Ground-Level Ultrasonic Sensors:** Placed closer to the ground for preliminary detection of objects on the ground. These sensors are chosen based on their precision in detecting non-cylindrical-shaped objects and insensitivity to changes in lighting.
- **Camera Module:** Placed for capturing images of detected objects. It helps the trained vision model differentiate real waste objects from non-waste objects. Added for balancing resolution with low power consumption.

Robotic Arm and Actuators

- **Servo Motors:** Installed in robot arm joints to ensure precision gripping and controlled movement. Selected for their good fine control ability and compactness.
- **End-Effector (Gripper):** Miniature mechanical gripper capable of lifting light weight waste items. Emphasis laid on toughness and flexibility to accommodate multiple shapes of waste.

Chassis and Support Frame

- **A light structure with structural brackets for the arm, sensors, and camera.** Selected to afford stability while the arm travels without inclining the mobile platform.

3.5.2 Computational support

A balance of onboard computation and cloud-based model training is required for decision making in garbage detection and collection.

Onboard Processing

- **ESP32 with Camera Interface:** Used as the main controller to insert ultrasonic inputs, drive servo motors, and send image data for processing. Chosen due to its dual-core capabilities, connectivity, and compactness for real-time applications.

Cloud and Backend

- Machine Learning Model: Trained waste classification model using labeled images of Plastic, Paper, Glass, and Polythene. Used in light-weight mode suitable for embedded inference.
- Backend Framework: Node.js and Express.js for storage of detection log, arm action data, and synchronizing with cloud services.
- Database: MongoDB for storing annotated dataset, detection outcome, and waste collection history.

Development Machines

- Standard laptops/PCs are used for dataset preparation, model training, and debugging. They provide sufficient computational support at reasonable cost without spending on high-end machines.

3.5.3 Supporting resources

For easy working, connectivity, power supply, and collaboration facilities are provided.

Communication

- Wi-Fi and Bluetooth connectivity to ESP32 for passing detection results and synchronizing classification outputs with the backend.

Power

- Rechargeable Li-ion battery packs employed to supply power to sensing modules as well as actuators of robotic arms. These are selected for high energy density and recyclability to allow long collection cycles.

Software Tools

- Arduino IDE: Used for programming microcontroller-based sensing and control code.
- Python & TensorFlow/PyTorch: Used for image pre-processing, training data sets, and exporting light-weight ML models to deploy.

Collaboration and Project Tools

- GitHub: To store collaborative code repositories and manage detection/classification logic.
- MS Planner: To manage milestones and organize dataset contribution effectively.

3.6 Data requirements

The waste detection and collection unit relies extensively on sensor- and vision-based data in order to detect and collect waste materials correctly. This data plan's purpose is to specify the kind of data required, how they will be obtained, and how they will facilitate the detection–collection process.

3.6.1 Object detection and recognition data

The primary task of this module is identifying garbage objects from the rest of the surrounding objects.

- Image data would be necessary for training and validation of the machine learning algorithm. Images of actual garbage in railway station settings would be gathered and labeled into four groups: plastic, paper, glass, and polythene.
- Additional data would be images taken under different light conditions (daytime, nighttime, shaded platforms) to provide strong generalization.
- During testing, the robot-mounted camera will keep snapping live frames that are matched against the learned model to verify detection accuracy.

3.6.2 Proximity and triggering data

- Proximity data to confirm the physical presence of the object is needed for the robotic arm to grasp an object.
- Ultrasonic sensor readings will provide proximity data to ensure that the object is within the reach range when detected.
- Data logs will be collected in tests in which objects are placed at different distances to ensure consistent triggering of the arm if only an item is in the prescribed pickup area.

3.6.3 Robotic arm operation data

The second fundamental data set is with regard to data related to the collection motion itself.

- Actuator response times, gripping success rates, and pickup/drop accuracy will be measured by repeated collection trials.
- Different types and sizes of waste objects will be employed to test the flexibility of the arm, while also validating its maximum lifting capacity of 2 kg to ensure safe and reliable operation under real-world conditions.

3.6.4 Data collection method

Both controlled laboratory testing and semi-realistic settings such as mock railway platforms will be used to collect data. Each experimental trial will provide a collection of sensor logs, image recognition results, and results from the operation of the arm. These data sets will then be processed in order to maximize the detection thresholds, improve the trained ML model, and achieve maximum synchronization among the collection and detection phases. The cycle of data gathering, testing, and optimization will continually operate in order to achieve effective garbage detection and autonomous collection under real operating conditions.

3.5 Gantt Chart

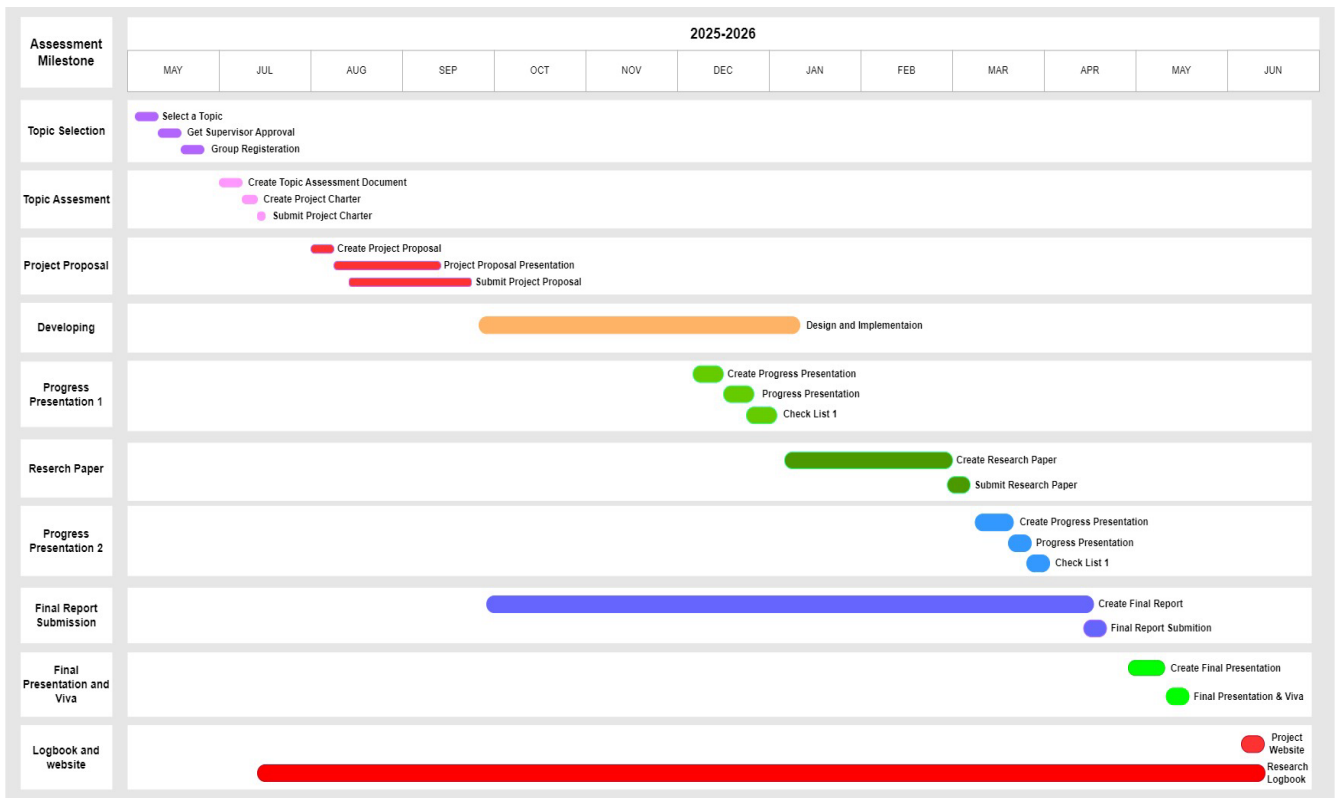


Figure 3.4: Gantt chart

3.6 Work Breakdown Chart

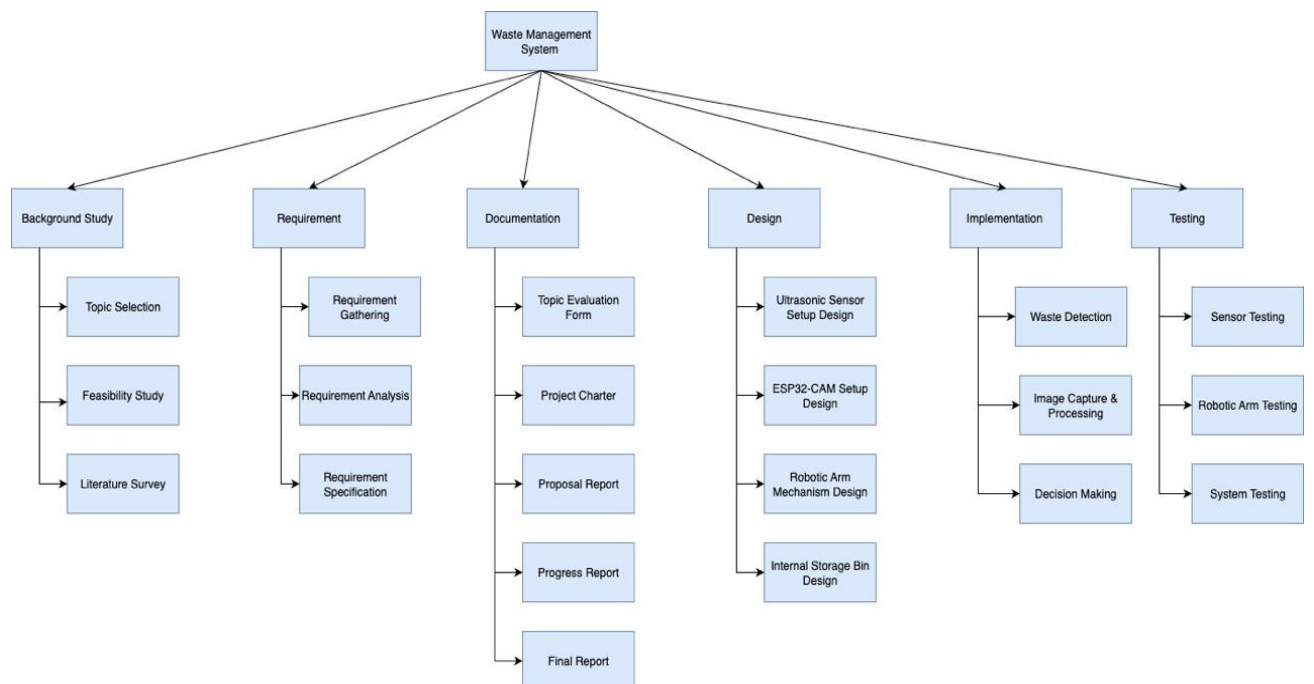


Figure 3.5: Work breakdown chart

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

- The robot should detect waste items in the railway station setting using onboard proximity and vision sensors.
- The system should sort detected items using the trained waste recognition model before collection.
- The robot arm should automatically pick up detected waste items and drop them into the internal bin.
- The detection and pickup should be a continuous process while in operation and not require human intervention.
- The system should handle various object sizes and positions in the robot arm's operating area.

4.2 Non-functional Requirements

- Performance: Detection and collection must occur in real time so as not to cause delays in the cleaning cycle.
- Accuracy: The waste detection model should have high recognition rates under different lighting and environmental conditions.
- Reliability: The robotic arm should consistently perform collection tasks with minimal failure rates.
- Scalability: The model must support retraining with new waste types if required in the future.
- Usability: The system should run independently without external controls, making it less complex to use for users.

4.3 User Requirements

- The robot must be able to detect and pick up waste independently without human intervention.

- The waste detection mechanism must have minimal false positives (e.g., not classify floor tiles or shadows as waste).
- The robotic arm must safely collect waste items without causing disruptions to passengers.
- The system should be capable of maintaining collection efficiency in dense locations without disrupting normal station operations.

4.4 System Requirements

Hardware:

- Image capturing and waste detection camera module.
- Distance measurement and item reachability confirmation ultrasonic/proximity sensors.
- Collection and bin placement robotic arm with servo/actuator-based joints.
- Temporary storage of collected items internal bin.

Software:

- Waste recognition custom-trained object detection/classification model.
- Sensor integration and decision-making embedded logic (detection → confirmation → arm activation).
- Control software for robotic arm movement and synchronization with detection results.
- Data logging and recording system for model accuracy and arm performance improvement over time.

5. DESCRIPTION OF PERSONAL AND FACILITIES

This component is managed by Perera W.B.N. with supervision from group supervisors. The tasks are developing the waste detection mechanism using sensors and a robotic arm for collection. They are supported with facilities such as an electronics lab, microcontrollers, vision modules, and prototyping space for testing the detection and collection process.

6. BUDGET AND BUDGET JUSTIFICATION

Table 6.1 Budget plan

Purpose	Estimated Cost (LKR)
ESP32 cam	1650
ESP32 cam adapter	500
Servo motor – 4 unit	3000
Robot arm frame integration materials (mounts, wiring, casing)	5000
Ultrasonic sensors (for detect grabage)	240
Total Estimated Cost	10390

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